Practical 7 and 8

Introduction

The COMPAS algorithm is a risk assessment tool used within the United States criminal justice system to estimate the likelihood that an individual will reoffend. Individuals are assigned a score between one and ten, with higher scores indicating a greater predicted risk of recidivism. Despite its widespread use, COMPAS has been the subject of significant academic and public debate due to concerns regarding bias and fairness. These concerns were first highlighted by investigative reporting and have since been examined in multiple peer reviewed studies (Dressel and Farid, 2018).

Methodology

This practical used the COMPAS two year recidivism dataset to examine fairness outcomes across demographic groups. Recidivism was defined using the two_year_recid variable. Predictions were treated as positive when an individual received a medium or high COMPAS score, defined as a decile score of five or above.

The analysis focused on group fairness for race and gender. Racial groups were categorised as Caucasian and non Caucasian, while gender groups were categorised as

male and female. For each group, confusion matrices were constructed and positive predictive value was calculated to assess predictive parity.

Results: Race

For non Caucasian individuals, the confusion matrix produced 1,530 true positives, 933 false positives, 1,542 true negatives and 755 false negatives. This resulted in a positive predictive value of 0.62.

For Caucasian individuals, the confusion matrix produced 505 true positives, 349 false positives, 1,139 true negatives and 461 false negatives. The resulting positive predictive value was 0.59.

Results: Gender

For male individuals, the confusion matrix showed 1,732 true positives, 994 false positives, 2,072 true negatives and 1,021 false negatives. This produced a positive predictive value of 0.64.

For female individuals, the results were 303 true positives, 288 false positives, 609 true negatives and 195 false negatives, resulting in a positive predictive value of 0.51.

Discussion

The results demonstrate that COMPAS predictions vary across demographic groups. While positive predictive value is relatively similar between Caucasian and non Caucasian individuals, more substantial disparities are observed in the gender based analysis. These findings reflect the wider fairness debate surrounding COMPAS, where predictive parity may be approximately satisfied for some groups while other fairness criteria, such as equalised odds, are violated.

This supports existing research which argues that it is not possible to satisfy all definitions of fairness simultaneously when base rates differ between groups (Dressel and Farid, 2018; Pessach and Shmueli, 2022). Studies have shown that optimising for one fairness

metric often introduces trade offs with others, particularly in high stakes domains such as criminal justice (Lagioia, Rovatsos and Prade, 2023).

Reflection

This practical highlights the ethical challenges associated with deploying algorithmic decision making systems in high-stakes environments. Even when protected attributes such as race or gender are not explicitly included in a model, biased outcomes can still arise due to correlations within the data (Pagano et al., 2023).

The COMPAS case demonstrates the importance of transparency, careful metric selection and ongoing evaluation. Fairness cannot be assessed using a single measure, and practitioners must remain aware of how different definitions of fairness can lead to conflicting conclusions. These findings reinforce the need for caution when using algorithmic tools in criminal justice, where decisions have significant consequences for individuals and communities.

References

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